

Data Dictionary

Term	Definition
Application	Refers to the “Calowin” mobile application.
User	The entity that is using the mobile application after registering for an account.
Account Database	Refers to the Account database of the “Calowin” mobile application, which stores all user-related information such as User ID, password, email. This database is separated from the App Database to enhance security within the application.
App Database	Refers to the App database of the “Calowin” mobile application, which stores all the information of the application. This database contains dynamic user data, including trip metrics (such as calories burned and carbon saved), and achievements belonging to the users. This data is updated in real-time based on user actions.
Google API	An Application Programming Interface which is used for map integration in the application.
NParks API	An Application Programming Interface which is called to extract various National Parks in Singapore.
Weather API	An Application Programming Interface which is called to retrieve live weather data and display in the application
Sign Up	Refers to a feature that allows the user to sign up for an account to use the features in the application.
One-time Password (OTP)	A si-digit combination code which is sent to the User via their registered email address. The OTP serves as an additional layer of security for the User in the scenario where they forget their login credentials.
Login	Refers to a feature which requires User to key in information consisting of an email, paired with a password which must be at least 8 characters and should include a combination of numbers, letters, at least 1 upper case and 1 lower case, and special characters that is created during sign up.
User Profile	Contains the user's basic information, including Name, userID, Bio, Weight and trip Achievement Badges.
Edit Profile	A feature which allows users to update their personal information and particulars such as their Name, Bio. Weight, and Password. User can also delete and wipe their account information through this feature.
Friends	Refers to other registered users of the application who are connected to a user’s Friendlist. Users can view their friends' achievements, stats and rankings in a leaderboard.

Friends List	A list which allows users to view their friends. Users can make a friend request to another registered user through the search of their userID. The friend list is accessible for all users using the app.
Leaderboard	A ranking system which ranks users based on their total high score which can be filtered by either 'Carbon Saved' or 'Calories Burned'. In the leaderboard, information such as points and achievement badges are displayed.
Achievement Badges	Badges awarded to users as rewards for various achievements. These achievements are unlocked by completing trips and progressing within the app.
Calories	A measurement of energy expenditure during physical activity. Calories burned are measured by the system based on the distance traveled as well as the mode of transportation used for the trip.
Carbon	A measurement of the reduction in carbon emissions when choosing eco-friendly transportation. Carbons saved are measured based on the distance traveled as well as mode of transportation used for the trip.
Trip	Refers to the user's movement from starting point A to destination point B.
Mode of Transports	Method of transportation in which one is using to travel from starting point A to destination point B, which includes car, walk, cycling and bus.
Trip Method	A feature of the application designed to allow users to plan and start their trips by selecting a preferred mode of transport. Based on the User's trips and selected transportation method, the application automatically calculates and displays the carbon savings and calories burned at the end of that trip.
Wellness Zones	Zones which contain green areas retrieved from NParks API such as parks and fields which are segmented based on certain regions of the island. User can toggle the radius between his/her location with the wellness zone and view the weather data corresponded to that zone.